Lab 2: Mobile Robot Motion and Control

Group 6
Abdalrahman Nasser - abdnas-2@student.ltu.se
Addo Dominic - domadd-2@student.ltu.se
Aittokallio Deborah - debait-2@student.ltu.se
Bardia Baigy - barbai-1@student.ltu.se

1. Objective

- **Describe the tasks and the goals. Summarize the lab in your own words.**

The objective of this laboratory was to gain hands-on experience with motion control of a mobile robot using the TurtleBot3 Burger platform. The lab focused on implementing and testing different control strategies in Simulink and ROS, including point-to-point navigation, trajectory following, line following, and wall following using LIDAR data.

- **What are potential difficulties? (Before you started)**

The potential difficulties that could arise for each task would be the development and tuning of the controllers. Additionally, another difficulty that could present itself would be the quality of the lidar data and connectivity used to achieve the goal of each task.

2. Methodology

- **What overarching method are you applying?**

A proportional control strategy based on the unicycle kinematic model was used throughout the lab. The robot's pose (x, y, θ) was obtained from odometry data, and control inputs were computed using position and heading errors. The unicycle kinematic model is given by:

- $\dot{x} = v \cos(\theta)$
- $\dot{y} = v \sin(\theta)$
- $\dot{\theta} = \omega$

where **v** is the linear velocity and **ω** is the angular velocity.

Position-Based Control

- Linear velocity was chosen proportional to the distance error to a target point (x^*,y^*):

$$\rho = \sqrt{(x^* - xo)^2 + (y^* - yo)^2}$$
$$v^* = Kv \cdot \rho$$

- The desired heading toward the target was computed using:

$$\theta^* = \tan^{-1}((y^* - yo) / (x^* - xo))$$

- Angular velocity (steering) was proportional to the heading error:

$$\gamma = Kh\ (\theta^* - \theta)$$

where **Kv** and **Kh** are positive control gains.

Trajectory Tracking

- Waypoints were used for trajectory tracking by applying the same point controller to a sequence of target points. The controller switched to the next waypoint when the distance error satisfied:

$$\rho < \varepsilon$$

Line and Wall Following

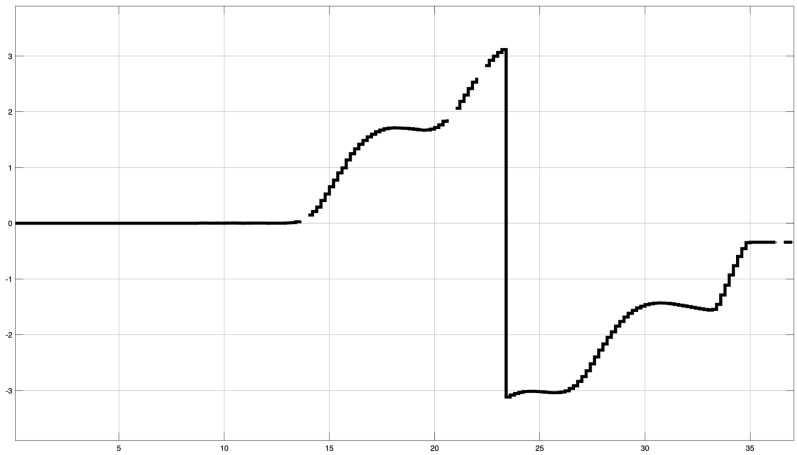
- LIDAR range data was processed to enable line-following and wall-following behaviors by computing distance and orientation errors relative to environmental features.

3. Task Breakdown and Results

Task 1: Position and Pose Control

The first task involved moving the robot from an initial position to a target position, and later to a full pose including orientation. The controller successfully reduced both position and orientation errors.

Plots:



- Orientation (θ) convergence plot

Task 2: Trajectory Following

Using the position controller, the robot was commanded to follow predefined trajectories including a square pattern, the group number, and a curved (sinusoidal) path. Waypoint switching was used to guide the robot along the trajectories.

Plots:

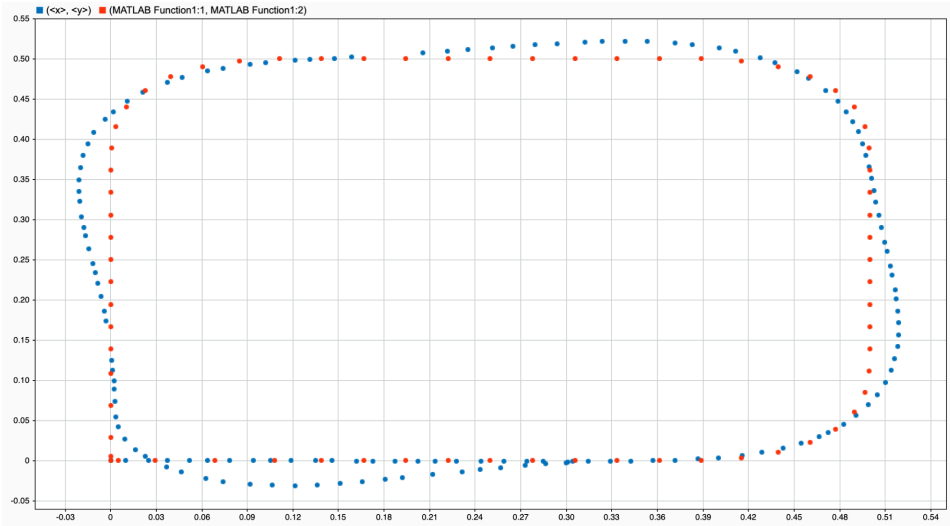


Fig. 1 Square trajectory (X–Y plot)

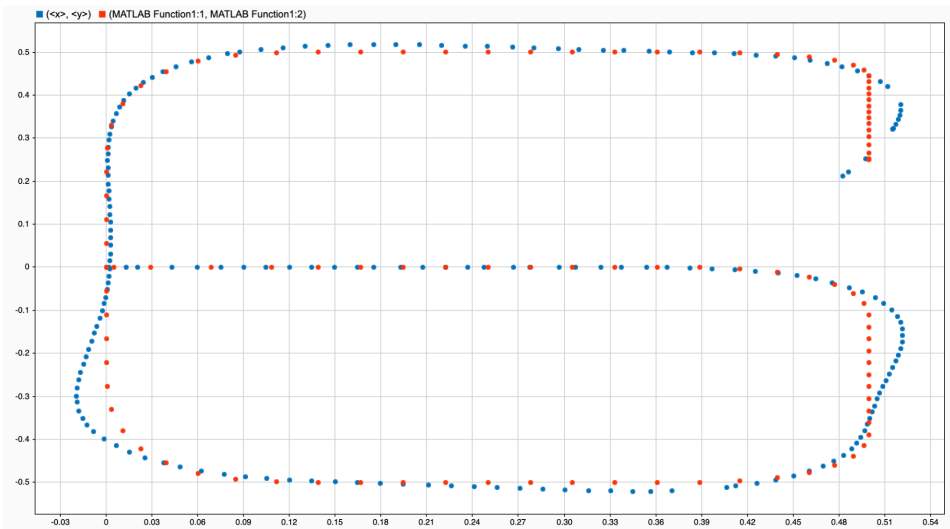


Fig. 2 Group number trajectory

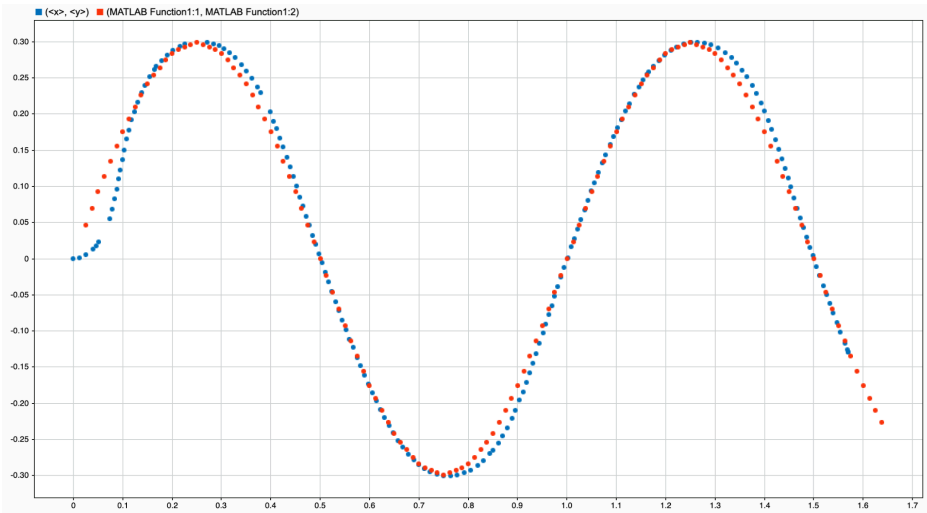


Fig. 3 Curved trajectory(Sinusoidal)

Task 3: Line Following

A controller was developed to follow lines defined by the equation (ax + by + c = 0). The robot adjusted its heading to minimize the perpendicular distance to the line and successfully converged to both tested lines.

Plots:

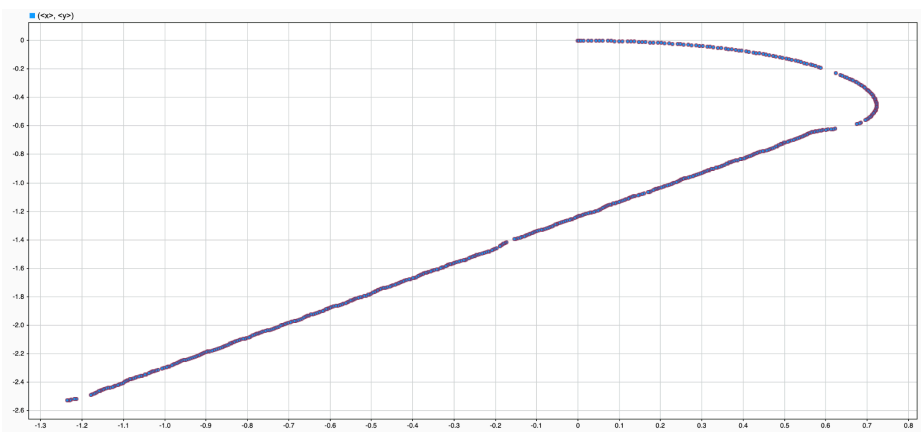


Fig. 4 Trajectory (Line 1)

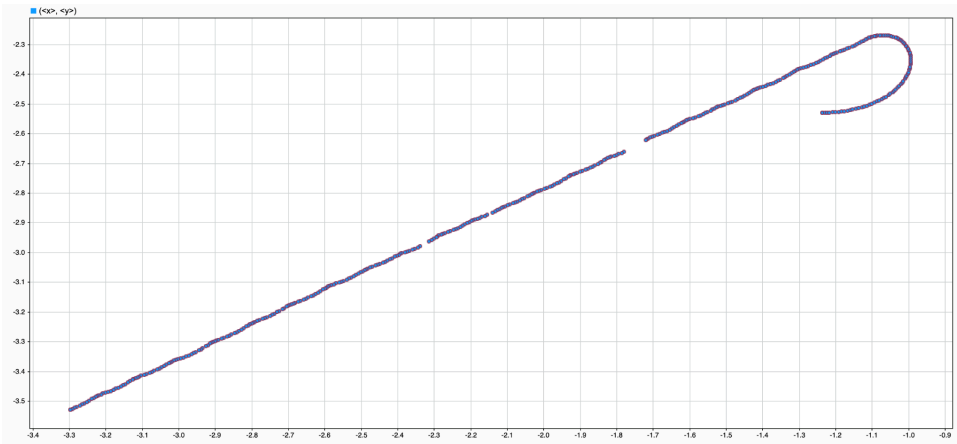


Fig. 5 Trajectory (Line 2)

Task 4: LIDAR Data Registration

LIDAR data was used to visualize the robot's surroundings. Individual scans were converted into Cartesian coordinates, and multiple scans were accumulated to generate a point cloud map. The effective minimum and maximum detection ranges were observed.

Plots:

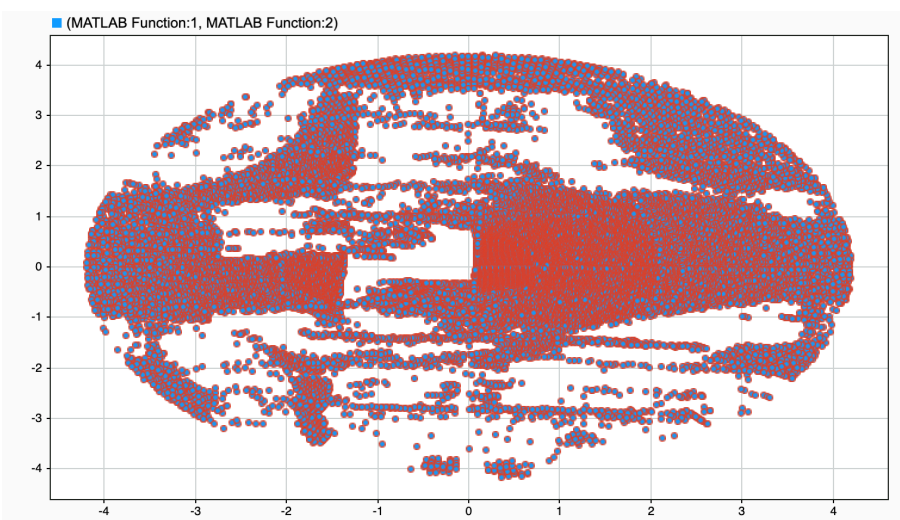


Fig. 6 LIDAR point plot

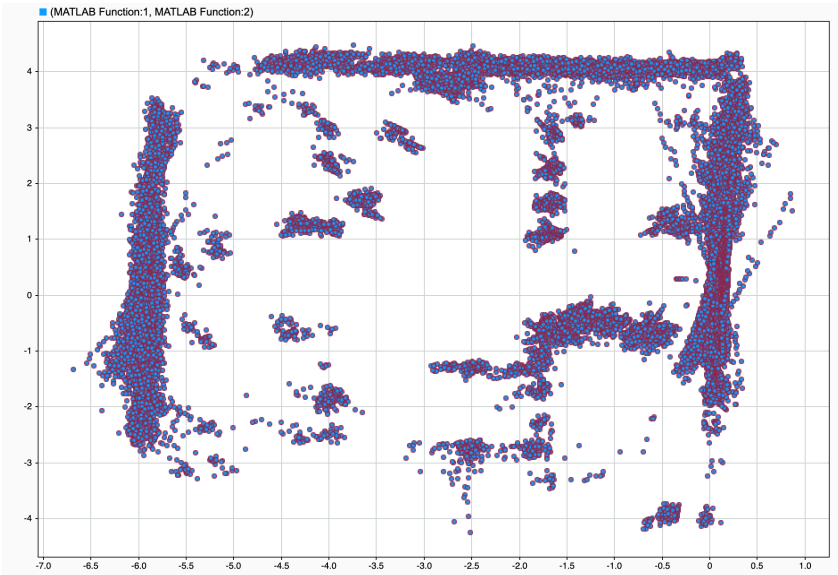


Fig. 7 Accumulated point cloud map

Task 5: Wall Following

The wall follower selects the closest valid LIDAR point on the right side, defines the current wall distance d_{curr} , computes the error `err =` d_{curr} - d_{ref}, and generates a lookahead target ("carrot") by moving L meters along the wall tangent while applying a normal correction proportional to err. This carrot point is then transformed from robot frame to world frame and passed to the position controller.

Screenshots / Plots:

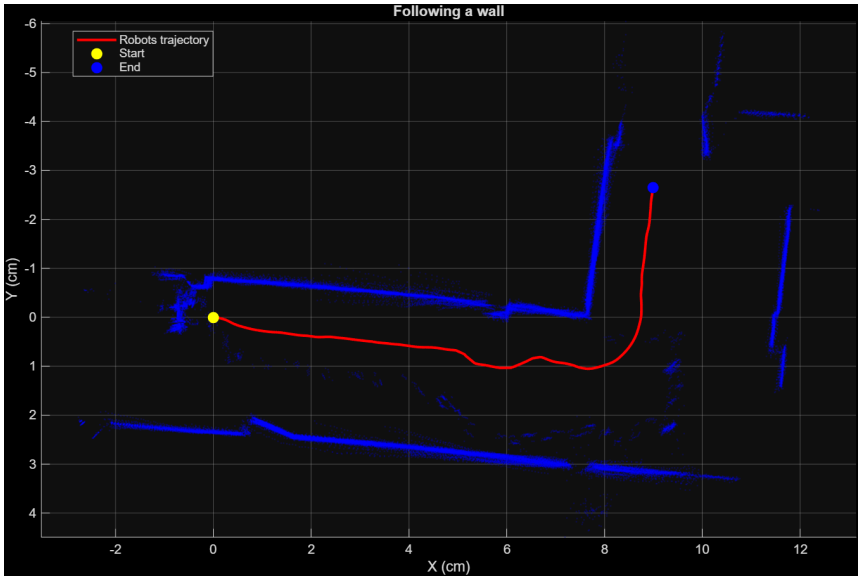


Fig. 8 Wall-following trajectory with the wall projected by LIDAR data

4. Difficulties

The main challenges involved tuning controller gains to avoid oscillations while maintaining responsiveness. Sensor noise from odometry and LIDAR also introduced small deviations, requiring careful thresholding and filtering.

5. What We Learned

This lab provided practical experience in mobile robot control, ROS-Simulink integration, and sensor-based navigation. We gained a deeper understanding of kinematic control, trajectory planning, and the use of LIDAR data for reactive behaviors in real robotic systems.